Assignment 5

Part 1: Memory Analysis

1. Check RAM and swap usage: free -h

2. Identify:

○ Total memory

○ Used memory

○ Available memory

○ Swap usage

```
ubuntu@ip-172-31-31-42:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           914Mi       356Mi       132Mi       2.7Mi       598Mi       558Mi
Swap:             0B          0B          0B
ubuntu@ip-172-31-31-42:~$
```

Total memory : 914MiB

Used: 356MiB

Available: 558MiB

Swap usage: 0B (swap memory is like virtual memory)

Part 2: System Statistics

1. Run: vmstat 5 5 (5 updates at 5 seconds delay)

2. Observe:

○ Memory usage

○ Swap in/out

○ CPU idle time

```
ubuntu@ip-172-31-31-42:~$ vmstat 5 5
procs -----------memory---------- ---swap-- -----io---- -system-- -------cpu-------
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
 2  0      0 135800   9504 603052    0    0    33    21   46    0  0  0 100  0  0  0
 0  0      0 135800   9504 603092    0    0     0     0   40   39  0  0 100  0  0  0
 0  0      0 135800   9504 603092    0    0     0     0   44   41  0  0 100  0  0  0
 0  0      0 135800   9504 603092    0    0     0     0   40   35  0  0 100  0  0  0
 0  0      0 135800   9504 603092    0    0     0     1   42   39  0  0 100  0  0  0
ubuntu@ip-172-31-31-42:~$
```

Memory:

free = 135800 KiB buff = 9504 KiB cache = 603092 KiB

Swap in/out:

swpd = 0 si = 0 so = 0. **No swap space in use** (swpd=0).

CPU idle time:

us = 0   sy = 0   id = 100   wa = 0   st = 0   gu = 0

- **id=100** on each sample ⇒ CPU is **~100% idle** (the machine is essentially doing nothing).
- us=0, sy=0 ⇒ no user/system CPU load.
- wa=0 ⇒ no I/O wait; st=0 ⇒ no CPU steal

Part 3: Load Average Interpretation

1. Run: uptime

2. Note the:

○ 1-minute

○ 5-minute

○ 15-minute load averages

```
ubuntu@ip-172-31-31-42:~$ uptime
 17:58:00 up  8:24,  2 users,  load average: 0.00, 0.00, 0.00
ubuntu@ip-172-31-31-42:~$
```

1-minute load → 0.00

5-minute load → 0.00

15-minute load → 0.00

This means that no processes are currently running in the CPU.

```
ubuntu@ip-172-31-31-42:~$ uptime
 17:58:00 up  8:24,  2 users,  load average: 0.00, 0.00, 0.00
ubuntu@ip-172-31-31-42:~$ yes > /dev/null &
[7] 7424
ubuntu@ip-172-31-31-42:~$ uptime
 18:07:33 up  8:33,  2 users,  load average: 0.08, 0.02, 0.01
ubuntu@ip-172-31-31-42:~$ uptime
 18:07:38 up  8:33,  2 users,  load average: 0.15, 0.03, 0.01
ubuntu@ip-172-31-31-42:~$ uptime
 18:07:41 up  8:33,  2 users,  load average: 0.22, 0.05, 0.02
ubuntu@ip-172-31-31-42:~$ uptime
 18:07:43 up  8:33,  2 users,  load average: 0.22, 0.05, 0.02
ubuntu@ip-172-31-31-42:~$ kill %1
```

/dev/null is a special file that throws away anything sent to it. So instead of printing "y y y y…" on the screen, all output is discarded. And we can see that this process is running in the background and simultaneously the cpu load has also increased.

4.

1. High load but low CPU usage → what could be the cause?

- It could be I/O wait, disk load, Network wait, Blocked processes where processes are waiting for something than CPU.

2. High swap usage → what does it indicate?

The system ran out of RAM and started using swap space

This indicates:

- Not enough physical RAM
- Too many memory-heavy processes
- If swap usage keeps increasing → system is struggling with memory.

3. When does adding RAM help vs optimizing processes?

RAM is added when there is insufficient memory space, which is a hardware upgrade. When we optimize the process, its because the load of CPU maybe higher, a memory leak might be present, too many unnecessary services running., which is a software solution.